FEA ANALYSIS OF A HIP IMPLANT



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1. Introduction:

A hip replacement is a type of surgery where a damaged hip joint is replaced with an artificial one, known as an implant. These are "designed to last for at least 15 years" [1] and "most people have a significant reduction in pain and improvement in their range of motion" [1]. In 2019, there were "81,130 hip replacement procedures" [2] with 93.8% of patients over 50 years old [2]; the highest rate of hip replacement was found to be in the 75-79 age group [2].

As the hip implants age, they will start to fail both due to internal fractures and also due to damage to the bones surrounding them. To check what state they're in, and to detect early-stage structural changes, vibration-based non-destructive testing (NDT) are used. These range from "radiography [which] can detect crack propagation or other complications" [3] to X-rays which are "another common imaging method" [3]. However, both of these have issues by both being potentially dangerous to patients or lacking the resolution needed to provide detailed information. The best method is therefore acoustic emission which "can be utilised to monitor both the initiation and development of material degradations" whilst giving detailed information on both the surface and internal failure modes of implants.

For this to work, the implant should have at least 5 natural frequencies within a certain frequency range of 100Hz to 3000Hz. This will be made out of the titanium alloy first to check whether the design meets the natural frequency and fatigue life criteria, and then compared to a stainless steel version. Once this has been done, an iteration will be made to the design to increase the number of natural frequencies within the given frequency range and tested for its fatigue life. Then, the same design will be tested using Stainless Steel to compare all the different models and conclude what the optimal design and material is.

2. Design, Assumptions & Properties:

2.1 The Original Design:

The original model [4] was found on GrabCad and was selected due to it both being accurate to the design but still simple enough for Ansys to run it without issues:

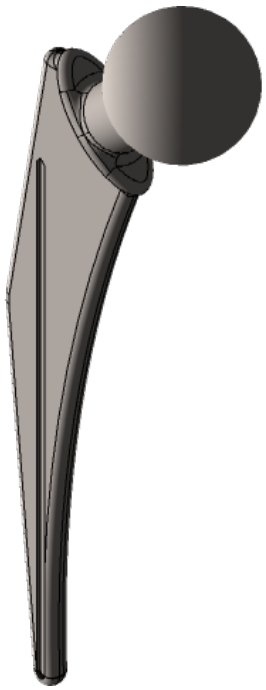


FIG 1: PICTURE OF THE ORIGINAL DESIGN

2.2 Assumptions:

From this design, assumptions had to be made regarding the geometry, materials and environment to get more accurate results and to simplify the model into something that could be run manageably:

- The whole components are made from one material and are a complete solid.
- The insert and cup are not modelled to avoid over-complexity.
- In order to accurately mimic the boundary conditions, as the implants are embedded in the femur, the hip implant's stem is only fixed from halfway down to its tail.
- In order to accurately mimic the daily loads, the implant's head is subjected to a vertical force which oscillates between 0 and 2,490 N. This is in the direction of the axis of the body.
- The materials are considered to behave linearly, so have linear elastic behaviour and infinitesimally small displacements and strains in the models.

2.3 Material Properties:

To be able to run the simulation accurately and calculate the correct natural frequency and fatigue life of the model, we needed to get an accurate S-N curve of the titanium alloy Ti-6AL-4V. This was found in a peer-review paper [5] and was then uploaded to Ansys in the Engineering Data to be used within the workbench and model:

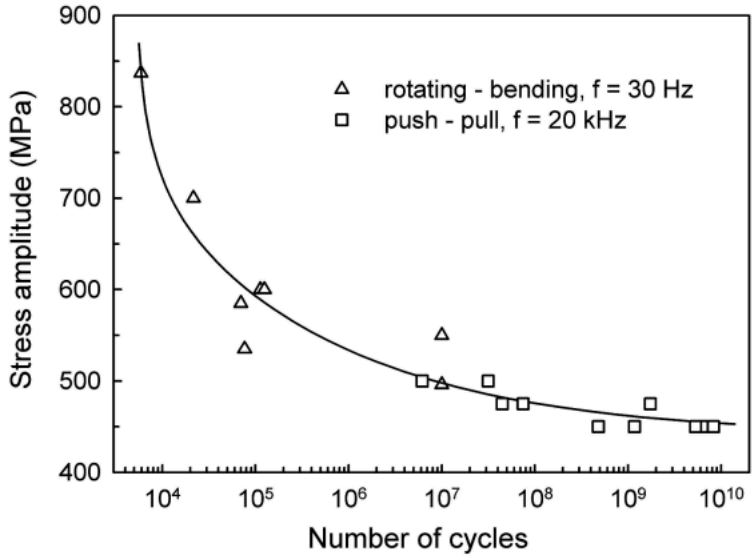


FIG 2: S-N CURVE FOR TI-6AL-4V [5]

For the Stainless Steel 316L instead, the S-N curve was taken from a different paper which investigated the fatigue behaviour of the material [6]. The S-N curve for this material had a "significant difference of fatigue life at high stress compared to lower stress" [6] which can be seen in the graph below:

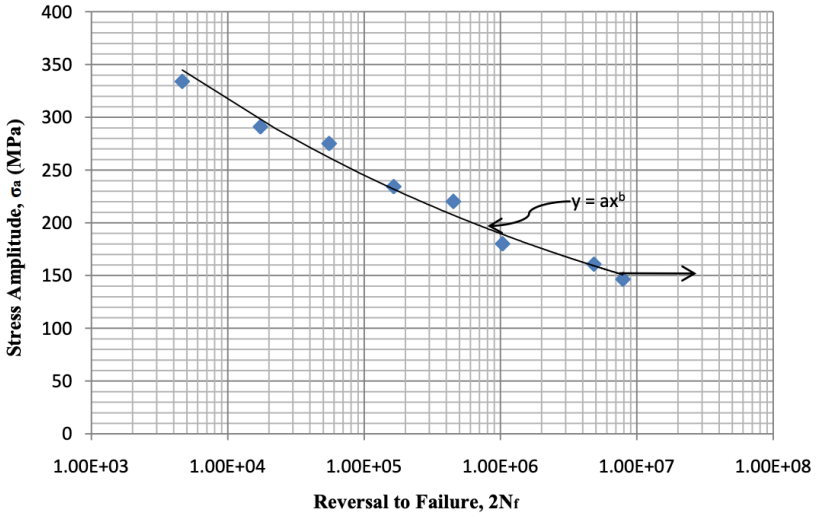


FIG 3: S-N CURVE FOR STAINLESS STEEL 316L [6]

The last bit of information we needed for both materials was the density, tensile strength and elastic modulus. These could be found online, using ASM [7] as a source for the Titanium and mat web [8] for the Stainless Steel. The table below shows these properties:

Material	Density (Kg/m³)	Tensile Strength (MPa)	Elastic Modulus (MPa)
Ti-6AL-4V	4430	950	113800
Stainless Steel 316L	8000	515	193000

Both materials offer, outside of these quantitive results, good corrosion resistance to "most chemicals, salts, and acids" [8] rendering them both extremely suitable to be used as implants. However, the obvious lightness of the Titanium compared to the stainless steel, being 45% lighter, whilst offering a higher tensile strength, makes it a more desirable material; with the only advantage of the steel being a lot cheaper and easier to manufacture.

3. Conditions:

3.1 Boundary Conditions:

As stated in the assumptions, the implant's stem is stated to be fixed from halfway down its length to its tail inside the femur. This is therefore modelled in Ansys as being a fixed support for that distance. This is the only fixed support as the head is assumed to be where the load is.

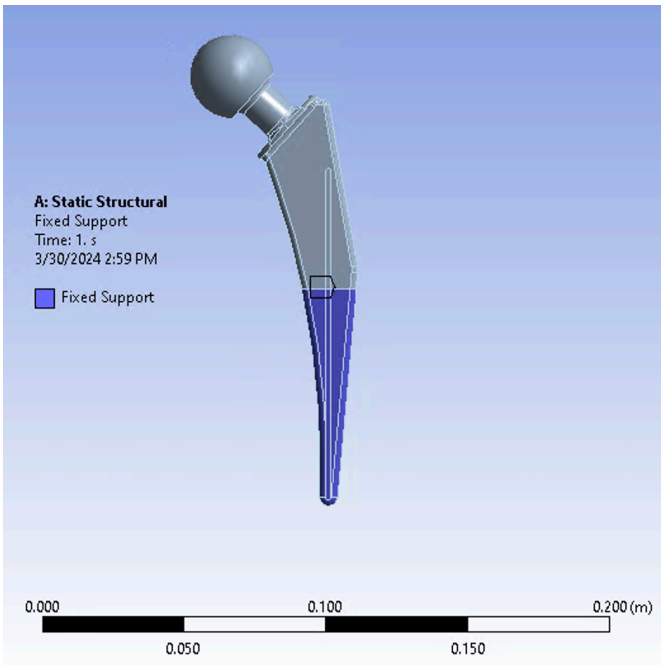


FIG 4. FIXED SUPPORT GEOMETRY

3.2 Loading Conditions:

The load is assumed to only act on the head, however, it is not constant for this model. Instead, this varies depending on the activity as the implant would be moving, thus, the force oscillates between 0 and **2,490 N** and acts vertically along the axis of the body; this load is thus known as the Live Load. We also have a Dead Load which acts on the body, being the weight of the parts of the body above the waist and divided in half due to there being two legs, therefore, this can be calculated to be a total of 51.46 Kg, or **504 N**.

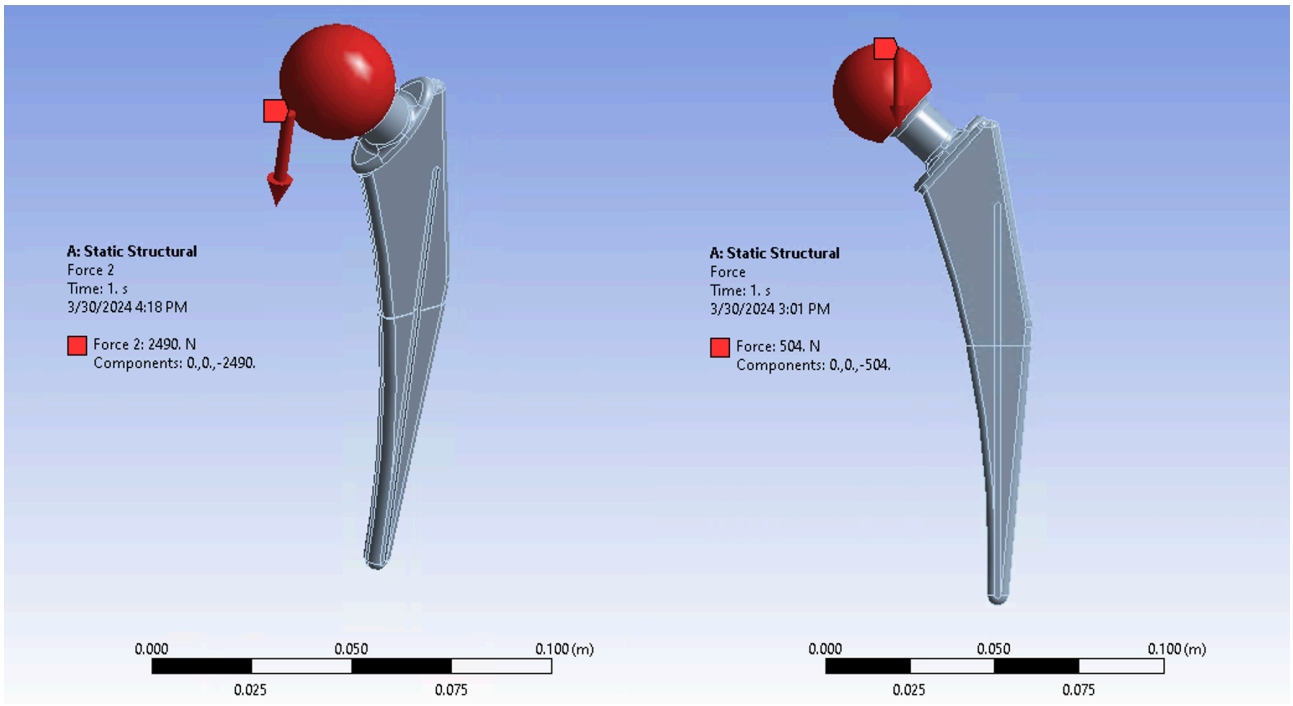


FIG 5 & 6. LIVE LOAD AND DEAD LOAD GEOMETRIES

4. Methods:

4.1 Natural Frequency Simulation Method

To find the fundamental frequency of the implant, a frequency simulation was run using the Modal setting. The objective was to check that the implant had at least **5** natural frequencies each within the range of **100 Hz** to **3000 Hz**, with changes that will be made to the design having the aim to increase this to more accurately give results on the condition of the implant when an AE test is run.

4.2 Fatigue Analysis Simulation Method

For this simulation, the objective was to work out if the implant can withstand an effective life cycle of 15 years, or 5 million loading cycles. This was with the loading conditions of a **504 N** dead load, and a live load that oscillates between **0** and **2,490 N**. To simulate this, a constant load of 504N was applied while the live load was run at max load to see if it could withstand it.

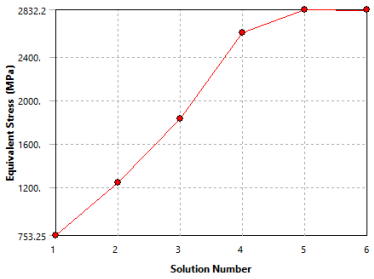
5. Sanity Checks:

In order to check that our assumptions and the model were correct to run accurate tests on, sanity checks were used to ensure that the output from both the modal and the static structural models made sense. The following sanity checks were implemented to check the results:

1. **Size and mass values:** The size of the model and its mass were compared to real-life examples [9] in order to check that the dimensions of all its sections were within the recommended range and thus accurate to a realistic model.
2. **Units used:** Within both the CAD software and Ansys, standard metric units were used to ensure consistency giving values in the mm, Kg, N, s, Mpa, etc.
3. **Material properties:** the material properties were taken from trusted sources and cross-compared to ensure that the model gave accurate results.
4. **Mesh refinement:** A mesh refinement study was run to check that the sizing was sufficiently low without increasing the computational time to such heights that it would be unfeasible over multiple iterations.

6. Mesh Refinement Study:

A mesh refinement study was run to ensure that two things. Firstly, the quality of the mesh should be as high as possible to ensure that the output from the model is as accurate as possible. However, we also want the computational time to be low and this increases as the accuracy of the mesh increases. Therefore, we use a mesh refinement study to get this delicate balance to be as optimised as possible. For our model, we compared the element size against the Von-Mises stress to find the best result and to see which of these settings converges:



	Equivalent Stress (MPa)	Change (%)	Nodes	Elements
1	753.25		48046	27243
2	1237.5	48.646	83943	51309
3	1828.4	38.552	153916	100067
4	2621.8	35.656	442074	303284
5	2829.4	7.6144	932007	654732
6	2832.2	9.8456e-002	1031706	725776

FIG 7. CONVERGENCE PLOT AND TABLE

When running mesh refinement, issues arose as we had Stress Singularities [10] within our model. To fix this, we refined the mesh locality in order to capture the wanted effect whilst also filleting the sharp corners as those often lead to stress concentrations. This led us to be able to run tests for convergence, giving us a refined mesh size of **1.25 mm**.

7. Results:

7.1 Titanium - Original design:

This design came out to weigh **0.115 kg**. Unfortunately, it only had 4 natural frequencies of frequencies ranging from **214 Hz** to **2448.2 Hz** meaning that it did not fulfil the requirements asked for.

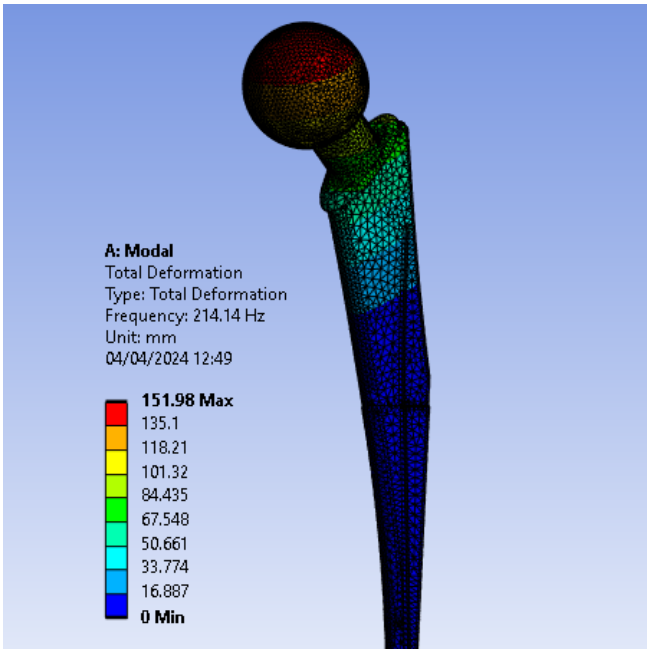


FIG 8. DEFORMATION AT THE FIRST NATURAL FREQUENCY

It could withstand a maximum Von-Mises Stress of **2960 MPa**, with a total maximum deformation of **1.05 mm**. However, the minimum amount of life cycles is 0 as it fails at the high stress point meaning that it cannot last the required 15 years.

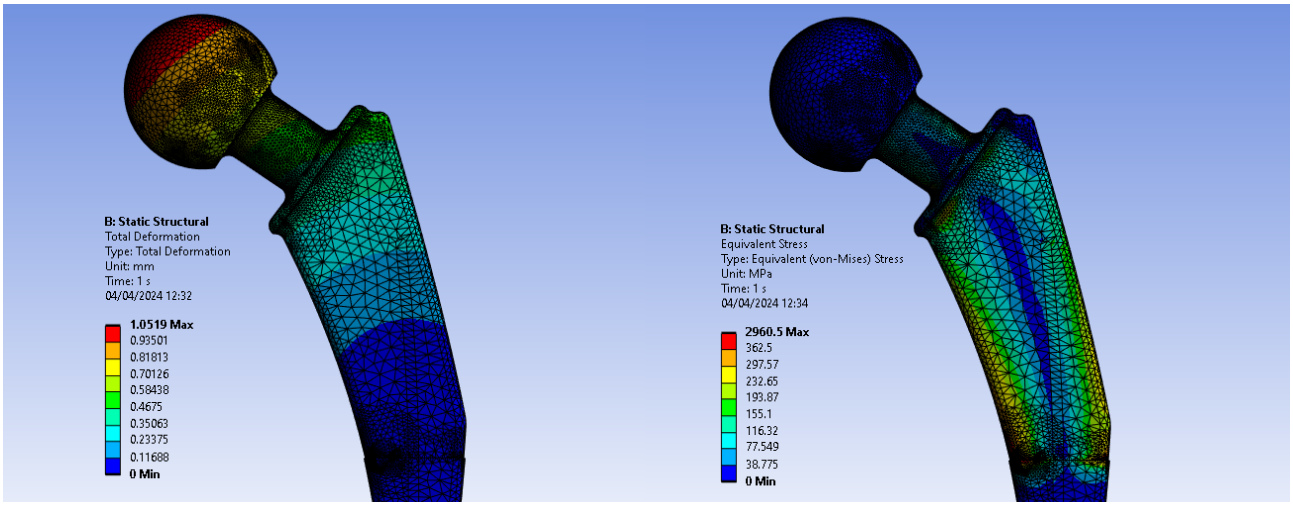


FIG 9 & 10. TOTAL DEFORMATION OF THE IMPLANT AND POINTS OF MAX STRESS

7.2 Stainless steel - Original design:

This version weighed **0.208 Kg**. Like the previous design, it only had 4 natural frequencies reaching a max of **2388.4 Hz** meaning it does not work for this application as well.

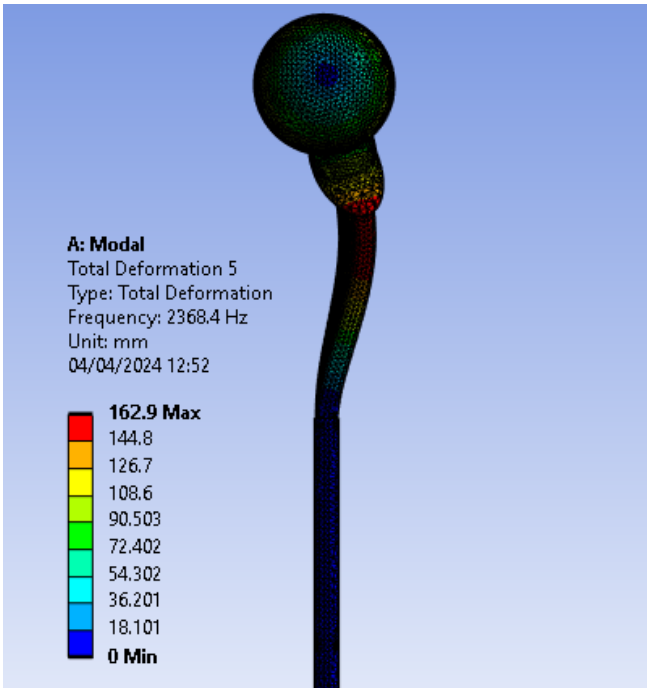


FIG 11. DEFORMATION AT 2385.1 HZ

With the stainless steel, the maximum Von-Mises stress became **2932.6 MPa**, with a total maximum deformation of **0.625 mm**. Unfortunately, steel cannot survive even one life cycle as it will fail in the red areas as shown in Fig 12.

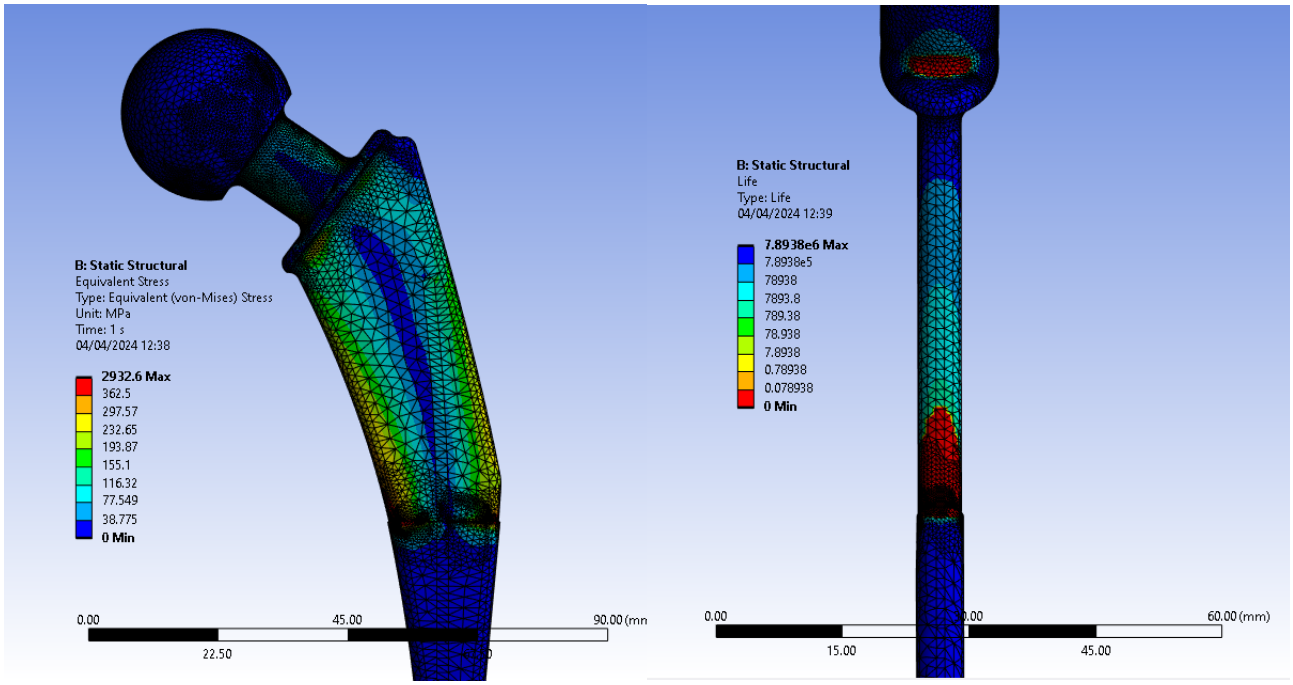
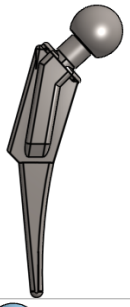
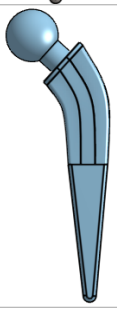
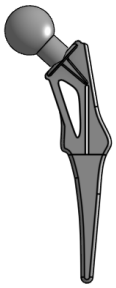
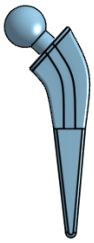
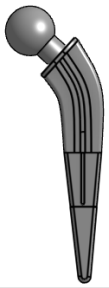
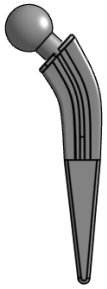
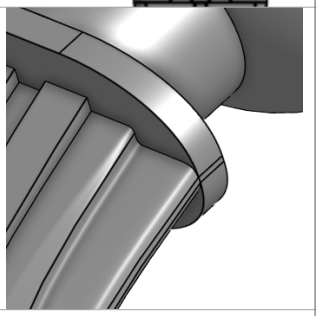
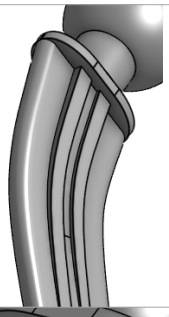
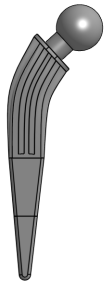
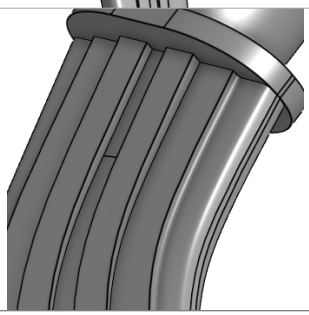
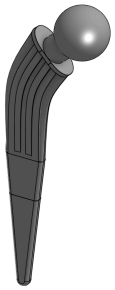


FIG 12 & 13. STRESS ON THE WHOLE IMPLANT AND POINTS OF MAX STRESS IN THE LIFE PLOT

8. Iterations:

V1: Added more support to stop forward / backward movement and side twisting. Natural Frequency was still too high.		V13 - V15: Played around with this design trying to get it to work. The main issue I was having was that I'd have really small points of very high stress that would fail in the life cycle. This could not be fixed by changing the mesh size so we went to a new design with smoother edges (V16)	
V2 - V3: Added holes to reduce mass and thus theoretically increase the natural frequency. Did not work well enough.		V16: A smooth shape was created in order to try and mitigate the issue I was having above of the small stress points. Whilst this worked for this design, the problem came back in later designs. This design almost had enough frequencies in range and did not fail the loading.	
V4: Stated experimenting with smoothing out the shape.		V17: Tried thickening the design to see what effects it would have. This unfortunately raised the natural frequencies but made it better for the loading.	
V5 - V6: Added another hole to decrease mass and tweaked its shape.		V18: Experimented with shelling the top head to reduce mass and so increase the number of low frequencies. This failed at loading.	
V7 - Tweaked it to the point where we got enough natural frequencies. However, it failed the life cycles.		V19: Added ribs to try to increase stiffness without increasing mass. This had a negative impact on the natural frequency so it was removed.	
V8 - V11: Played around and tweaked the smooth shape more. **The issue with getting the requirements is that to increase the natural frequencies, you need to reduce mass or make it stiffer, but reducing the mass means reducing the strength so it fails the life cycles.**		V20: Added an indentation to the sloped support and the fixed support at the bottom to reduce mass, thus improving natural frequency. This had an impact but it was not big enough to bring the values into the desired range.	

V21: Experimented with reducing the height of the stem in order to make the part of the implant that vibrates longer to improve the natural frequencies.		V29: thinned out the indentation to the point it was about to fail the life cycles.	
V22 - V23: Made the stem longer in order to have a similar effect to V21 without reducing the height of the fixed stem.		V30: Optimised the shape from the previous steps and smoothed it out until it was perfect.	
V24 - V25: Looking at the way the implant deformed due to the natural frequencies, I noticed that in the 5th one it was being twisted sideways. In order to make this easier I made the sideways supports thinner.		V31: Experimented again with the V21 changes.: This brought the natural frequency within the correct requirements but failed at the life cycles.	
V26 - V28: In order to increase the effect I was aiming for in V24, I decided to make an indent at the ends of the stem thinning it out. This was tweaked over these versions to get the best result for both frequency and life cycles.		V32: Tweaked the V31 changes to get the support to be as high as possible whilst having 5 frequencies within the 3000 Hz range and still surviving the life cycle.	

9. Iteration Results:

9.1 Titanium - Iterated Design:

The final iterated design came out to be **V32**. This is the only one that we got to have both 5 natural frequencies in the 100 to 3000 Hz range and to survive over 5 million loading cycles.

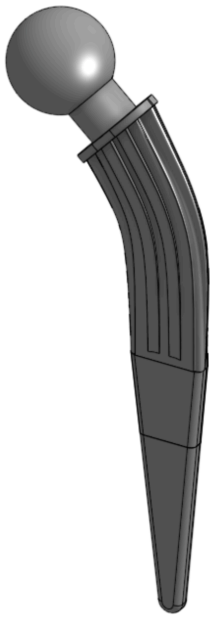


FIG 14. FINAL ITERATED DESIGN

For this design, the final weight was **0.130 Kg** and there were **5** natural frequencies in the 100 to 3000 Hz range; these being at: **141.16 Hz, 515.07 Hz, 977.15 Hz, 1800.5 Hz** and **2988.2 Hz**. From them, we got a total deformation plot with a max deformation of **138.45 mm**:

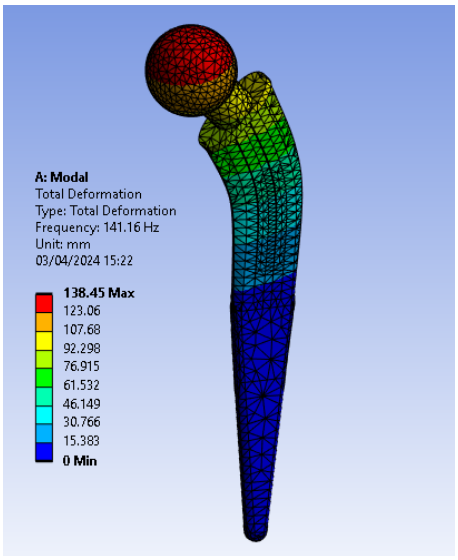


FIG 15. TOTAL DEFORMATION AT 141.16 HZ WITH TITANIUM

Meanwhile, the maximum Von-Mises stress was **2832.2 MPa** with an average of **144.86 MPa**. Fortunately, this design was able to survive **5.06 million** life cycles, which is just above the required 5 million threshold for its expected life of 15 years.

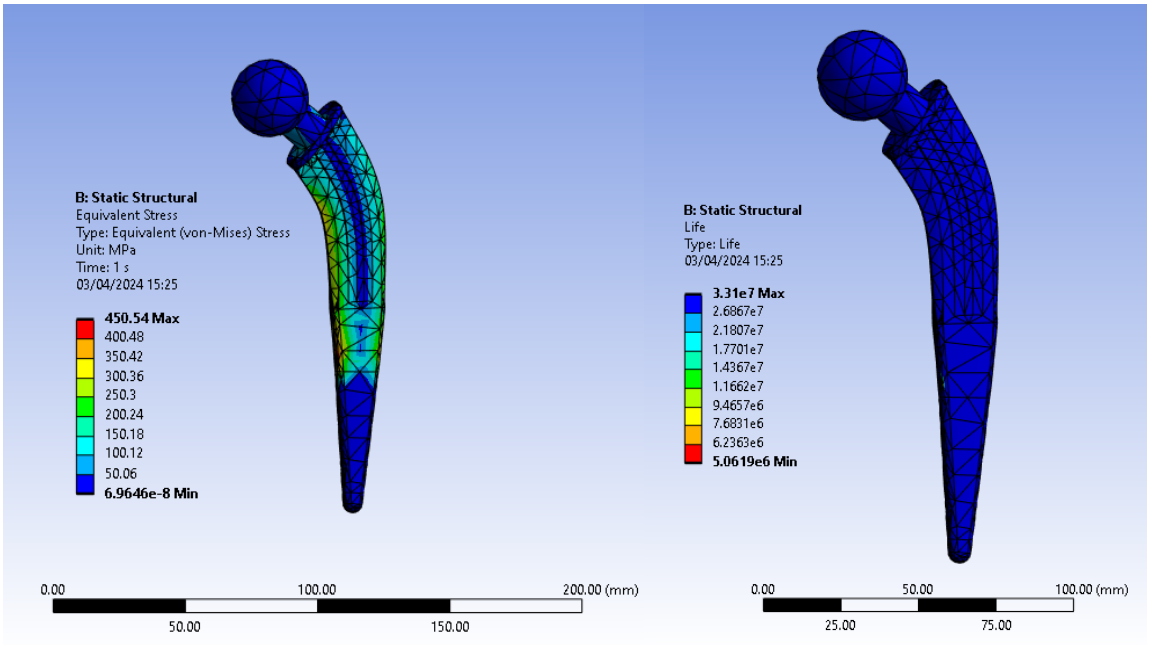


FIG 16 & 17. VON-MISES STRESS AND LIFE CYCLES FOR THE ITERATED DESIGN WITH TITANIUM

9.2 Stainless Steel - Iterated design:

With the stainless steel, the design weighted **0.234 Kg** but it still had 5 frequencies in the required range with a comparable max frequency of **2890.9 Hz** and **134.76 mm** of max deformation. However, when it came to surviving the life cycles, the design unfortunately failed at **0** cycles with a max stress of **829.31 Mpa**.

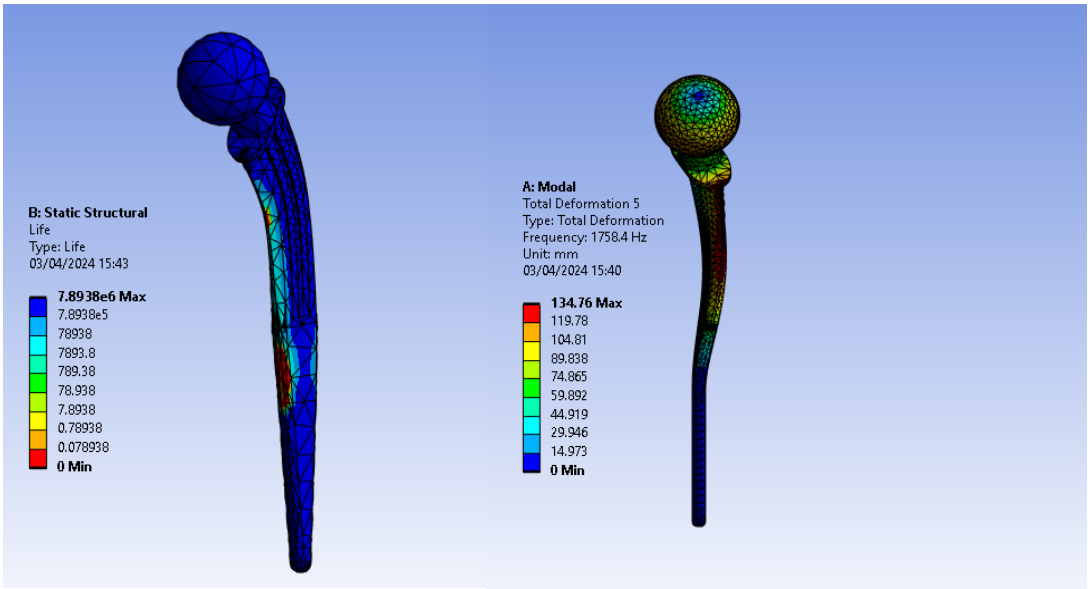


FIG 18 & 19. LIFE CYCLES AND TOTAL DEFORMATION FOR THE ITERATED DESIGN WITH STAINLESS STEEL

10. Discussion & Conclusion:

Many issues arose with trying to fulfil the given design requirements. Both the initial design and multiple iterations were needed to be able to get the design to a point where it could survive the 5 million loading cycles over its 15-year life and be able to have 5 natural frequencies within the range. The problem that arose was that the model needed to have a fine balance between the stiffness and strength that is required for it to survive loading, and the opposing lack of it and mass that is needed for the implant to be able to resonate to get the required natural frequencies.



FIG 20. RENDER OF THE HIP IMPLANT

By the last iteration, using the titanium alloy, the implant was able to theoretically survive 5.06 million life cycles bringing it just about above the required threshold for the lifetime of the device. I was also able to get it to have 5 natural frequencies in the 100 to 3000 Hz range meaning that a doctor could run acoustic emission on it to test for micro fractures and cracks within the metal that could have developed from the loading and unloading of the implant. However, to ensure both of these theoretical results apply to the real world, practical testing should be carried out on the device to ensure that they reflect and work in the real world and through whatever extreme scenarios it might be subjected to such as during intense work outs.

11. References:

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